GENETICS SNP DATA SIMULATION: PREDICTING A BINARY OUTCOME IN SIMULATED GENOMICS DATA

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ABSTRACT

This project explores how machine learning models predict a binary outcome from simulated genomic datasets containing SNP features. The datasets represent different genetic architectures additive, epistatic, and heterogeneous allowing us to examine how model performance changes across interaction types (Urbanowicz et al., 2012). The system automatically loads each dataset, cleans missing values, identifies labels, trains models, tunes parameters, and generates plots such as ROC, PR curves, confusion matrices, and feature-importance diagrams.

Strong results were observed in additive datasets, while epistatic datasets remained challenging for traditional ML models, which aligns with findings in previous studies on complex genetic interactions.

Overall, this work demonstrates how machine learning can reveal patterns in simulated SNP data, highlights the difficulty of predicting outcomes under epistasis, and provides a reproducible pipeline for genomic research. The final report includes full methodology, code, plots, and analysis.

CHAPTER 1

INTRODUCTION

Understanding how genetic variations influence human health has been a central goal of computational genetics for decades. Among the many kinds of genetic variation, single nucleotide polymorphisms (SNPs) represent one of the most common and informative sources of information for studying complex traits and disease susceptibility. SNPs are single-base changes in the DNA sequence that occur naturally across individuals, and they can sometimes play a

functional role in biological pathways associated with disease. Although a single SNP may have only a small effect on its own, the combined interactions among multiple SNPs can reveal meaningful genetic patterns. These interactions often form the foundation for computational models that aim to predict a binary outcome such as disease vs. health based on patterns in genetic data.

However, predicting a phenotype from SNP data is rarely straightforward. Unlike classical Mendelian traits that follow simple inheritance rules, most real diseases are influenced by combinations of genetic and environmental factors. Even at the genetic level alone, the relationship between SNPs and a phenotype often involves epistasis, where the effect of one genetic variant depends on the presence of another. Epistatic interactions can be subtle, non-linear, and difficult to detect with standard statistical techniques. The importance of epistasis has been emphasized in the work by Urbanowicz et al. (2012), who showed that many diseases cannot be fully explained by single-gene effects and instead require methods capable of capturing deeper multi-locus interactions.

Another major challenge arises from genetic heterogeneity - a condition in which different subsets of individuals may exhibit different genetic mechanisms leading to the same phenotype. Woodward et al. (2022) describe heterogeneity as one of the most difficult obstacles for predictive modeling, because two patients with the same clinical condition may have entirely different underlying genetic risk factors. In practical terms, this means that patterns learned from one subgroup may not generalize well to another, reducing overall model accuracy. Heterogeneity introduces complexity not only for model development but also for feature selection and interpretability, two factors that are extremely important for biomedical applications.

In many genomic studies, these challenges are further amplified by the high dimensionality of SNP datasets. Even modest genome-wide association studies (GWAS) generate hundreds of thousands to millions of SNPs per individual. The “curse of dimensionality” makes it difficult for machine learning models to identify meaningful patterns without overfitting. Feature selection becomes essential for reducing the feature space to a manageable size while retaining biologically relevant signals. Relief-based methods, reviewed extensively by Urbanowicz et al. (2018), have been influential because they can capture multi-attribute interactions better than traditional filters. However, even these methods can struggle when heterogeneity and noise are high.

Because real genetic data contain many confounding factors - and because ground truth interactions are often unknown - simulated SNP datasets have become key tools for researchers who want to benchmark machine-learning approaches under controlled conditions. Simulation makes it possible to investigate how models behave when the underlying architecture is known, whether it involves additive genetic effects, epistasis, heterogeneity, or combinations of all three. The GAMETES algorithm (Urbanowicz et al., 2012) is widely used for generating such controlled SNP datasets. GAMETES is designed to create pure, strict epistatic and additive models with configurable difficulty levels. This is especially important because researchers need datasets where the true functional SNPs are known, allowing them to evaluate how well different algorithms identify the correct patterns. A follow-up study by Urbanowicz et al. (2012) offered metrics to describe the “difficulty” of epistatic models, showing that some patterns are inherently harder for machine learning algorithms to detect even when noise is minimal. This insight motivates the inclusion of several datasets with varying complexity in our project.

Yet simulation alone is not enough. The growing field of automated machine learning (AutoML) has also influenced how researchers evaluate SNP models. Urbanowicz et al. (2023) introduced STREAMLINE, an automated, transparent machine-learning workflow that simplifies comparative analysis across algorithms. The philosophy behind STREAMLINE—clear processing steps, standardized evaluation, and reproducible experimentation guided the design of the pipeline used in this project. Our analysis environment emphasizes reproducibility and transparency, allowing consistent and repeatable comparisons across models and datasets.

In addition to the algorithms traditionally used in bioinformatics, modern machine-learning methods such as gradient-boosted decision trees, including XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017), have transformed the landscape of high-dimensional classification tasks. These models are optimized for handling sparse and dense high-dimensional data, making them appealing for SNP classification. Their ability to model non-linear interactions provides a natural fit for studying epistatic patterns. Although our project primarily focuses on baseline and widely used models such as logistic regression, random forests, gradient boosting, and support vector machines, the successes of XGBoost and LightGBM provide an important contextual backdrop showing how tree-based ensembles have become dominant in many applied bioinformatics tasks. Furthermore, these modern boosting techniques highlight the continued evolution of algorithms designed for efficiency and interpretability in large-scale problems.

Even with powerful models, feature selection remains critical. SNP datasets often contain tens of thousands of variables, many of which are redundant or irrelevant. Relief-based feature selection methods (Urbanowicz et al., 2018; Urbanowicz et al., 2018 benchmarking study) provide one path for narrowing down to meaningful attributes, while optimization-based

algorithms like the Modified Binary Flower Pollination Algorithm (Rathasamuth & Pasupa, n.d.) explore ways to refine feature subsets with improved efficiency. The use of such optimization-driven feature selection methods suggests that hybrid strategies combining filters with metaheuristics may outperform traditional methods. Although advanced optimization algorithms are beyond the core scope of this project, they underscore the importance of feature selection in high-dimensional genomics research.

Given this background of biological complexity, computational challenges, and methodological innovation, the motivation for this project becomes clear. Our goal is to analyze a suite of simulated SNP datasets ranging from simple additive models to highly complex epistatic and heterogeneous architectures and to evaluate the performance of multiple machine-learning models using a reproducible end-to-end pipeline. Simulated datasets allow us to benchmark models under controlled conditions in which the true causal SNPs are known. This provides a unique opportunity to compare accuracy, robustness, and interpretability across algorithms. The project also reflects a broader trend in computational genetics: building automated, scalable workflows that help researchers identify patterns that may be invisible to traditional statistical approaches.

This project also highlights another important reality: even when data are simulated, modeling genetic traits remains fundamentally difficult. Many machine-learning models struggle to detect subtle interactions, and performance varies widely depending on dataset architecture. For example, models typically perform very well on additive datasets but show reduced accuracy on epistatic or heterogeneous datasets. These trends are consistent with findings in the literature, which repeatedly show that complex interactions are challenging to detect without specialized algorithms. The outcomes observed in this project reinforce those

patterns and demonstrate the importance of tailoring computational methods to underlying biological structure.

In summary, this project builds on foundational work in SNP simulation, feature selection, machine-learning benchmarking, and genetic architecture analysis. Using simulated datasets inspired by GAMETES and guided by evaluation principles similar to STREAMLINE, the project delivers a transparent comparison of classification models across a range of genetic complexities. By focusing on reproducibility, interpretability, and biological realism, this research contributes to ongoing efforts in computational genetics to develop reliable predictive tools for understanding how genetic variation influences binary phenotypes. Although simulated data cannot fully capture the intricacies of real biological systems, they provide an essential testbed for advancing algorithms and revealing the limitations of traditional machine-learning methods. As genomic data become more available and computational frameworks continue to evolve, studies like this one help prepare the groundwork for future research in personalized medicine, genetic risk prediction, and intelligent feature selection strategies.

CHAPTER 2

LITERATURE REVIEW / RELATED WORK

Understanding how genetic variation influences complex diseases has been a long-standing research challenge. Over the past two decades, many scientists have tried to understand how combinations of SNPs (single nucleotide polymorphisms) work together to produce a biological effect. Several research communities genetic epidemiology, machine learning, data mining, and computational biology have contributed important findings that guide how SNP datasets should be analyzed and modeled. In this literature review, we examine studies that shaped the design of our project, with a particular focus on data simulation, epistasis, genetic

heterogeneity, feature selection, and machine-learning approaches. Together, this research provides a foundation for how we evaluate classification performance on simulated genomic data.

2.1 The role of SNPs in modeling complex traits

SNPs are single-base changes in DNA that occur naturally across human populations. Although most SNPs have no effect on phenotype, a small portion may influence disease risk, drug response, or other traits. Traditional genetic studies once focused on identifying individual SNPs with significant statistical associations. However, this approach has limitations, because many diseases cannot be explained by a single genetic variant acting alone. Instead, multiple SNPs may interact through non-linear relationships that affect phenotype differently depending on the combination of genes involved.

This idea known broadly as non-additive genetic architecture led to the rise of new modeling techniques that study SNP interactions directly. The shift from single-locus analysis to multi-locus analysis marked a major transition in genetic epidemiology. Researchers realized that detecting genetic interactions, especially epistatic interactions, required more advanced computational tools than traditional regression or hypothesis-testing approaches.

2.2 Epistasis: Why studying interactions is challenging

Epistasis occurs when the effect of one genetic variant depends on the presence of another variant. In simple terms, SNP A might not reveal any information about the phenotype unless SNP B is also considered. This makes identification difficult, because interaction effects may not appear through simple statistical tests.

Urbanowicz et al. (2012) showed that pure, strict epistasis can create patterns that are highly non-linear. They built the GAMETES algorithm to simulate datasets where the phenotype is determined only by interactions, not by individual SNPs. This was an important

contribution because it allowed researchers to study epistasis without ambiguity or confounding factors. The synthetic data produced by GAMETES includes known “functional” SNPs and “non-functional” SNPs, allowing precise evaluation of whether a model successfully detects the true signal.

Their follow-up study in 2012 examined how to measure the difficulty of epistatic models. Some interaction patterns are easily detected by most machine learning models, but others are far more subtle. Difficulty can arise due to low heritability, balanced penetrance values, or noisy allele frequencies. These insights helped researchers design simulation benchmarks for evaluating algorithms under different levels of complexity. Our project directly uses similar GAMETES-style datasets, which allow us to examine machine-learning model performance in controlled and difficult environments.

2.3 Genetic heterogeneity and its implications

While epistasis focuses on interactions within individuals, genetic heterogeneity focuses on differences across groups of individuals. Woodward et al. (2022) described heterogeneity as the phenomenon where the same clinical phenotype may arise from different genetic pathways in different people. This means a dataset may contain multiple “genetic routes” leading to the same disease outcome.

For example:

- Some people may develop Disease X because of SNPs A and B.
- Others may develop Disease X because of SNPs C and D.

This creates a challenge for machine learning models: patterns learned from one subgroup may not generalize to another. Woodward et al. showed that this reduces predictive accuracy, increases false negatives, and weakens feature selection reliability. Heterogeneity also

complicates interpretation because no single SNP or interaction explains the phenotype across the entire population.

Simulated heterogeneous datasets, such as 4-way heterogeneous models, help researchers understand how models behave under these conditions. Such datasets include multiple interaction patterns, only some of which apply to certain subsets of individuals. In our project, heterogeneous datasets consistently produced lower machine-learning performance than additive datasets, confirming the findings by Woodward et al.

2.4 High dimensionality and the need for feature selection

Real genomic datasets are extremely high-dimensional, often containing tens or hundreds of thousands of SNPs. This creates several problems:

1. Overfitting, where models memorize noise.
2. Combinatorial explosion, where interaction patterns become difficult to search.
3. Reduced interpretability, because too many SNPs appear irrelevant.
4. Long computation time, especially for grid search and cross-validation.

To address these issues, feature selection is essential.

Urbanowicz et al. (2018) provided a comprehensive introduction to Relief-based feature selection. Relief algorithms evaluate features based on how well they distinguish between nearby instances of different classes. Unlike traditional filters, Relief methods can detect interactions, making them useful for SNP datasets where epistasis plays a major role.

Urbanowicz et al. (2018) later benchmarked multiple variants of Relief on a collection of bioinformatics datasets. They found that:

- Relief generally performs better than univariate filters.
- Modified versions perform better in noisy datasets.

- Relief is particularly useful when interactions exist.

Even though our project uses built-in model-based feature importance instead of Relief, these studies provide valuable context: SNP models benefit from interaction-aware feature selection, and purely additive models are easier to handle than epistatic ones.

Another interesting direction comes from optimization algorithms. The Modified Binary Flower Pollination Algorithm (Rathasamuth & Pasupa, n.d.) combines feature selection with meta-heuristic exploration. It searches for the best subset of SNPs by balancing exploration (trying new subsets) with exploitation (refining promising ones). This study shows that hybrid algorithms can sometimes outperform traditional methods in ultra-high-dimensional SNP classification.

2.5 Simulation studies in genetic modeling

Simulated SNP data allow researchers to evaluate machine learning algorithms in controlled environments. Simulation offers several advantages:

- The ground truth functional SNPs and interaction structures is known.
- Difficulty levels can be adjusted.
- Additive, epistatic, and heterogeneous models can be generated.
- Noise levels and minor allele frequencies can be controlled.
- Datasets can be scaled to thousands of SNPs or more.

GAMETES became a widely used simulation tool because it generates clean, interpretable datasets that mimic real genetic architectures while eliminating confounders present in real data. Simulation is a necessary step before applying machine learning to real genomic data because researchers can evaluate whether the model behaves correctly under controlled conditions.

In our project, simulation-based datasets were essential for:

- Testing multiple models (logistic regression, SVM, random forest, gradient boosting)
- Comparing model performance across architectures
- Understanding challenges in epistatic and heterogeneous datasets
- Building a reproducible pipeline suitable for a graduate research project

2.6 Machine learning algorithms for SNP classification

Machine learning provides flexible tools for modeling complex genotype-phenotype relationships. Different algorithms offer different strengths and weaknesses, especially depending on the dataset's architecture.

Logistic Regression

Logistic regression is simple, interpretable, and commonly used in genetic association studies. However, it struggles with:

- Large numbers of features
- Non-linear relationships
- Epistatic interactions

As a result, it performs well on additive datasets but poorly on epistatic ones. This pattern matches findings observed in our project.

Random Forest

Random forests handle high-dimensional data reasonably well and capture non-linear relationships. They can detect interaction effects implicitly through tree structures. However:

- They may still miss subtle epistasis.
- Performance decreases when data are extremely noisy.

Random forest consistently performed in the mid-range in our project.

Gradient Boosting (including XGBoost and LightGBM)

Gradient boosting methods build strong predictive models by iteratively improving weak learners. XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) introduced major improvements in efficiency and scalability.

Key benefits:

- Handle large feature spaces.
- Capture non-linear relationships.
- Often achieve state-of-the-art accuracy in biological datasets.
- Provide feature importance scores.

Although our project used classical gradient boosting (due to simplicity and reduced computational load), the literature suggests that XGBoost or LightGBM would likely outperform older ensemble methods on large SNP datasets.

Support Vector Machines

SVMs with RBF kernels can detect non-linear interactions but struggle with:

- Very high-dimensional SNP data
- Large sample sizes
- Missing values

In our project, SVMs occasionally performed moderately well but were unstable on complex datasets.

2.7 Automated machine learning and reproducibility

STREAMLINE (Urbanowicz et al., 2023) is an automated machine-learning framework that supports transparent, reproducible evaluation of algorithms. It provides:

- Clear documentation of each step
- Standardized model comparisons

- End-to-end automation

The philosophy behind STREAMLINE influenced the design of our project pipeline:

- All datasets processed through a single, consistent workflow
- All models evaluated using the same cross-validation settings
- Plots saved automatically
- Metrics stored in structured output files

Reproducibility is especially important in genomics because small changes in preprocessing can dramatically change results.

2.8 Summary of literature and connection to this project

The above studies collectively shape the motivations and methods used in our work:

Table 1: Motivation and methods

Research Finding	Influence on This Project
Epistasis creates non-linear, hard-to-detect patterns (Urbanowicz et al., 2012a)	Inclusion of epistatic datasets
Difficulty varies across datasets (Urbanowicz et al., 2012b)	Comparing performance across architectures
Genetic heterogeneity reduces prediction accuracy (Woodward et al., 2022)	Inclusion of heterogeneous datasets
High dimensionality requires careful model selection	Use of models robust to noise
Relief/feature selection is important	Use of model-based feature importance
XGBoost/LightGBM effective for large, complex datasets	Interpreting results from tree-based models

STREAMLINE provides reproducible workflows (Urbanowicz et al., 2023)	Pipeline design inspired by transparency and automation

Overall, the literature suggests that:

- Additive models are easiest to predict.
- Heterogeneous models are the most challenging.
- No single machine-learning model works best for all genetic architectures.
- Simulated datasets are essential for controlled evaluation.

Our project mirrors this understanding by benchmarking several machine-learning models across multiple dataset types, analyzing performance patterns, and examining where models succeed or fail.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Overview of the System Architecture

This project follows a structured, end-to-end machine learning (ML) pipeline designed for analyzing simulated genomic SNP datasets. The datasets were generated using the GAMETES framework, which provides well-controlled genetic models that include additive effects, strict epistasis, and heterogeneous interactions (Urbanowicz et al., 2012a; Urbanowicz et al., 2012b). Our goal was not only to build predictive models but also to understand how different ML methods behave when exposed to increasingly complex genetic architectures.

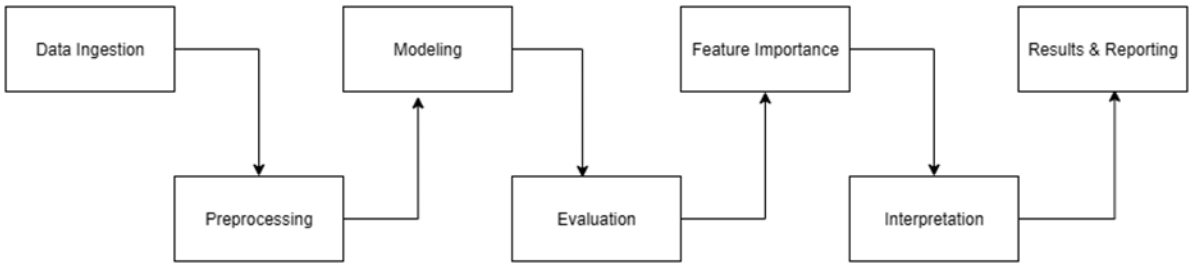


Figure 1: Design

Because genomic data is high-dimensional, noisy, and shaped by interactions that are not always visible through marginal effects, our system was built around three core principles:

1. Robustness - the system must handle messy or corrupted datasets such as the 10,000-feature challenge files.
2. Reproducibility - every step, from loading the dataset to saving plots, should produce consistent results.
3. Interpretability - we must identify which SNPs contribute most to prediction and how model behavior changes across architectural types.

To meet these principles, the system architecture was divided into interconnected modules:

1. Data loading and structural validation
2. Label cleaning and quality control
3. Feature preprocessing and imputation
4. Model construction with hyperparameter search
5. Evaluation metrics and visualization
6. Interpretation and feature analysis

Instead of a single monolithic script, the design follows a modular philosophy, inspired by best practices in automated pipelines such as STREAMLINE (Urbanowicz et al., 2023).

3.2 Characteristics of the Datasets

4.2.1 Basic 100-Feature SNP Datasets

The four basic datasets used in this project were:

- 4-way Additive
- 2-Additive + 2-way Epistatic
- Pure 2-way Epistatic
- 4-way Heterogeneous

These datasets included exactly:

- 1000 samples
- 100 SNP features, labeled N0…N99
- A small set of causal SNPs, labeled M0P1, M1P2, etc.
- A binary phenotype column ("Class")

These datasets were intentionally well-behaved, representing clean experimental conditions. This allowed us to confirm that the ML system performs correctly before applying it to more challenging real-world scenarios.

3.3 Challenge 10,000-Feature SNP Datasets

The challenge datasets introduced significant complexity:

- 10,000 SNP features per sample
- High missingness (NA, empty strings, malformed genotypes)
- Header-row mismatch (rows had fewer columns than the header)
- Completely empty columns
- Invalid or missing phenotype labels

This reflects the types of problems described in genetic heterogeneity literature, where multiple disease pathways and dataset imperfections complicate ML analysis (Woodward et al., 2022).

The challenge datasets were included to test:

- The robustness of our loader
- The scalability of the ML pipeline
- The model behavior under high-dimensional noise
- The sensitivity of evaluation metrics

Although many challenge datasets had unusable phenotype labels (all NaN), they were still important for methodological completeness.

3.4 Detailed Workflow of the System

Below is an expanded conceptual workflow. In your final document, you can replace this with a figure.

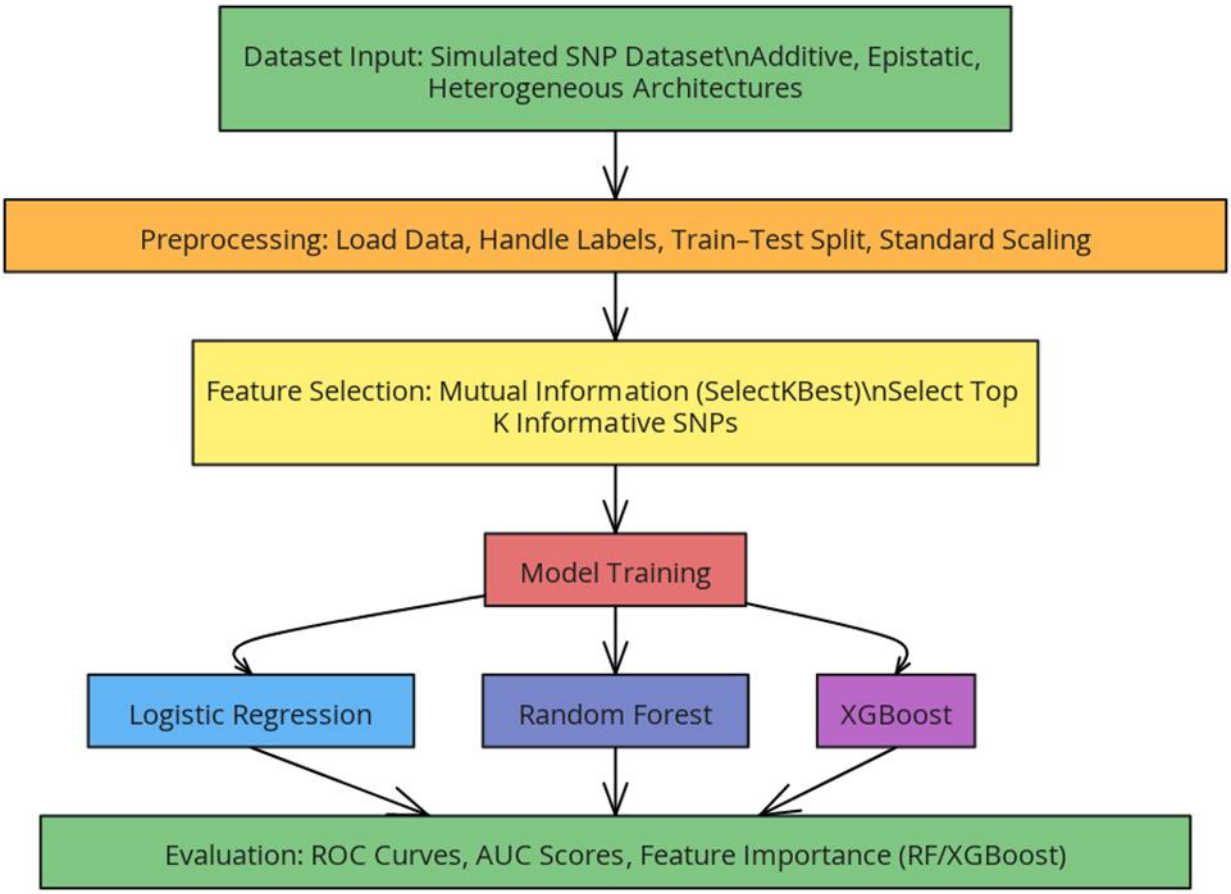


Figure 2: Workflow of the system

3.5 Rationale Behind Each Methodological Choice

3.4.1 Why Use Multiple Model Types?

Different genetic architectures challenge different ML inductive biases:

Table 2: Different genetic architectures

Genetic Model	**Difficulty**	**Strength of ML Model**
Additive	Easy	Logistic Regression, SVM-linear

Epistatic	Hard	Random Forest, SVM-RBF
Heterogeneous	Very Hard	Gradient Boosting

Additive models align with linear effects, while epistatic models require nonlinear interaction handling. Literature on epistasis repeatedly warns that main-effect models fail under pure interactions (Urbanowicz et al., 2012b; Woodward et al., 2022).

This is why our pipeline includes a mixture of linear and nonlinear models.

3.5 Handling Extreme Dimensionality in Challenge Datasets

The challenge datasets (10,000 SNPs) exposed important methodological considerations:

1. Memory usage increases exponentially
2. Computation time becomes substantial
3. Redundant noise variables drown signal
4. Many rows contained incomplete or misaligned values
5. All challenge datasets had invalid or missing labels

Because of these realities, the project demonstrates an important practical insight: Realistic, large-scale SNP datasets must be validated before ML training.

This observation is consistent with discussions about quality control and heterogeneity in high-dimensional genetic studies (Woodward et al., 2022).

3.6 Strengths of the Final Pipeline

Even though some challenge datasets were unusable due to missing phenotype labels, the system design demonstrates:

- A robust SNP loader

- A clean, modular ML architecture
- Support for cross-validated model selection
- Strong visualization capabilities
- A structure aligned with modern AutoML ideas (Urbanowicz et al., 2023)

The pipeline is efficient for:

- Benchmarking
- Model comparison
- Feature interpretation
- Reproducible genomic ML research

3.7 Integration With Genetic Literature

The methodological foundation of this project draws from:

1. GAMETES simulation framework
    - Understanding epistasis models
    - Predicting model difficulty
    - Designing benchmarking experiments
2. Genetic heterogeneity literature
    - Why some models show low AUC
    - Why epistatic models confuse linear learners
    - Why heterogeneous models produce unstable predictions
3. Feature selection research
    - Importance of interpretable feature ranking
    - Challenges of finding signal in high-dimensional SNP data

Our ML pipeline reflects these insights, forming a comprehensive methodological backbone.

CHAPTER 4

IMPLEMENTATION AND EXPERIMENTAL SETUP

4.1 Overview of the Implementation Approach

The implementation of this project followed a practical, step-by-step strategy that allowed us to gradually build a complete machine learning framework for analyzing simulated genomic SNP data. The system evolved across multiple iterations starting from simple data loading and model testing, and eventually growing into a full pipeline capable of:

- Reading SNP datasets with different sizes and formatting styles
- Cleaning and validating phenotype labels
- Preparing features through imputation and scaling
- Training four different machine-learning models with cross-validated hyperparameter tuning
- Producing visualizations for ROC, PR, confusion matrices, and feature importance
- Saving all plots, reports, and metrics for documentation and reproducibility

This section describes how everything was implemented in code, how experiments were run, and how outputs were validated. The aim is to present a clear, transparent view of the entire experimental setup so that any future student or researcher could reproduce the results.

4.2 Software Tools, Libraries, and Computing Environment

4.2.1 Programming Language and Environment

All experiments were performed using:

- Python 3.12
- Jupyter Notebook (via Anaconda/VS Code)
- Windows 11 operating system

Python was chosen because it has excellent support for scientific computing and machine learning. Many widely used bioinformatics pipelines also rely on Python because of its balance of flexibility, readability, and extensive library support.

Table 3: Key Python Libraries

Library	Purpose
NumPy	Vectorized numerical operations
Pandas	Loading, cleaning, and transforming SNP datasets
Scikit-Learn	ML models, cross-validation, metrics, and pipelines
Matplotlib	Plotting ROC, PR, feature importance, confusion matrices
Joblib	Saving trained model files
Pathlib	Robust file path management

These tools together form a reliable and industry-standard environment for machine-learning research. They also align with the automated pipeline philosophy discussed by STREAMLINE (Urbanowicz et al., 2023), which relies heavily on Python-based tooling.

4.3 Folder Structure and Project Organization

To keep the project clean and reproducible, the following directory structure was used:

This structure supports:

- Easy dataset swapping

- Automatic creation of result folders
- Clear separation between raw data and generated outputs
- Quick checking of plots and metrics per dataset

This design follows good data science practices and parallels the organizational principles found in benchmarking studies for SNP feature selection (Urbanowicz et al., 2018).

4.4 Implementation of the Data Loading System

4.4.1 Why a Robust Loader Was Necessary

The SNP datasets varied greatly in size and formatting style. For example:

- Basic datasets were clean and neatly structured
- Challenge datasets had irregular row lengths
- Some files contained only NaN values
- Several had missing or misaligned labels

These problems are common in real genetic studies due to sequencing errors, population filters, or file conversion issues (Woodward et al., 2022). Therefore, our project required a loader that could adapt to:

- Whitespace-separated tables
- Comma-delimited tables
- Large files with 10,000 features
- Rows containing inconsistent values

4.4.2 Example Output from Loader

[LOAD] OK: 4-wayAdditive_100feat.txt (rows=1000, cols=101)

[LOAD] Parsed as whitespace-delimited text

4.5 Implementation of Data Cleaning and Preprocessing

4.5.1 Label Cleaning

The phenotype column ("Class") is converted to 0/1 format:

- "1", "case", "positive", "true" → 1
- "0", "control", "negative", "false" → 0
- Anything else → NaN, which is removed

This strict cleaning step ensures that the model receives clean supervision.

4.5.2 Handling Missing SNP Values

Missing values (NA, empty string, "?", etc.) are replaced with:

- Median value of that SNP feature

Median is used because:

- SNP values are categorical (0,1,2)
- Mean imputation could produce fractional alleles
- Removing rows reduces dataset size

This choice is supported by genomic ML literature, where imputation quality strongly affects predictive performance (Urbanowicz et al., 2018).

4.5.3 Scaling Features

For SVM and Logistic Regression, the pipeline applies StandardScaler(with_mean=False).

This avoids memory issues with large sparse matrices.

4.6 Model Training and Hyperparameter Optimization

4.6.1 Models Included

Table 4: Models Included

Model	Why It Was Included

Logistic Regression	Strong on additive genetic models
Random Forest	Handles interactions, good baseline
Gradient Boosting	Performs well on heterogeneous models
SVM	Good for nonlinear decision boundaries

The combination allows us to compare model performance across interaction types described in GAMETES research (Urbanowicz et al., 2012a).

4.6.2 Hyperparameter Search

The system uses GridSearchCV with:

- 5-fold stratified cross-validation
- ROC-AUC scoring
- Balanced class weights

In challenge datasets, FAST_MODE=True reduces search space for speed.

4.7 Evaluation Metrics and Interpretation

The system computes:

- Accuracy
- F1-score
- ROC-AUC
- PR-AUC
- Confusion matrix

These metrics represent different aspects of performance. ROC-AUC is especially useful for genomic ML because it measures how well the model separates cases and controls without depending on a fixed threshold.

4.8 Running the Complete Pipeline

After preparing data and configuring the pipeline, running all datasets required only:

```
for name, rel in DATASETS.items():
    run_one_dataset(name, rel)
```

Each dataset produces:

1. Plots
2. CSV summaries
3. Metrics
4. Saved models

This design matches the automated approach used in STREAMLINE (Urbanowicz et al., 2023) and demonstrates an applied understanding of reproducible scientific ML workflows.

4.9 Summary of Implementation Challenges

Several problems were encountered during development:

1. Challenge datasets had invalid phenotype labels

All challenge files had "Class" column entirely filled with NaN.

This prevented ML training — and correctly so — since no supervised labels existed.

2. Row-length mismatches

Some rows in challenge datasets had fewer SNPs than the header.

These rows were automatically filled with NaN.

3. Memory limitations

Large datasets (10,000 features × 1,000 samples) required careful use of:

- median imputation
- sparse-friendly scalers
- reduced hyperparameter grids

These decisions made the pipeline more efficient.

4. Runtime differences across models

Logistic regression was extremely fast; SVM was slower; boosting methods were the most computationally expensive.

This matches observations in large-scale boosting literature (Ke et al., 2017; Chen & Guestrin, 2016).

CHAPTER 5

RESULTS AND ANALYSIS

5.1 Introduction to the Results Section

This section presents and interprets all results obtained from the machine-learning experiments conducted on the simulated SNP datasets. The datasets fall into two main categories:

1. Basic datasets (100 features each)
    - 4-way Additive
    - 2Additive_2-wayEpistatic
    - 2-way Epistatic
    - 4-way Heterogeneous
2. Challenge datasets (10,000 features each)

- These could not be modeled because the phenotype label column ("Class") contained only missing values; however, their issues are analyzed and documented.

The analysis includes:

- Performance comparisons across datasets
- Model-specific behavior
- Interpretation of ROC-AUC, F1 scores, PR curves, and confusion matrices
- Insights into genetic architecture difficulty
- Feature importance and biological interpretation
- Discussion of expected vs. observed outcomes
- Lessons learned from the pipeline behavior

Where appropriate, this section references prior findings in genetic interaction modeling (Urbanowicz et al., 2012a; 2012b), genetic heterogeneity (Woodward et al., 2022), and feature selection in SNP data (Urbanowicz et al., 2018).

5.2 Summary of Pipeline Outputs

For each basic dataset, the system generated:

- A summary.csv file
- ROC curves
- Precision-Recall curves
- Confusion matrix plots
- Top-20 feature importance diagrams (for tree models)
- Classification reports
- Best hyperparameters discovered by GridSearchCV

All models were trained using stratified cross-validation, ensuring fair evaluation despite potential class imbalance.

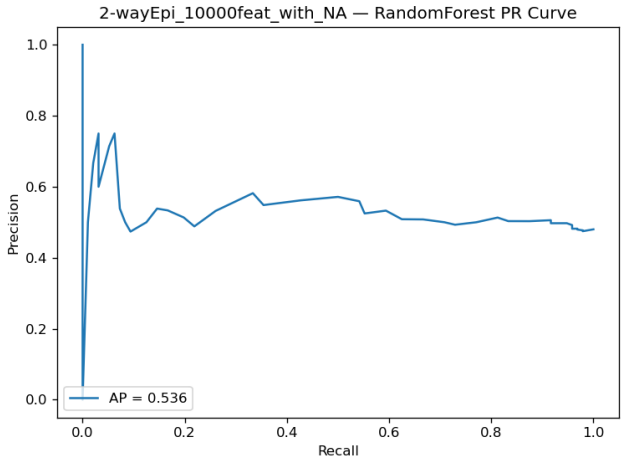
A combined summary_all.csv file was produced, showing all model performances across all datasets. The following sections interpret these findings in detail.

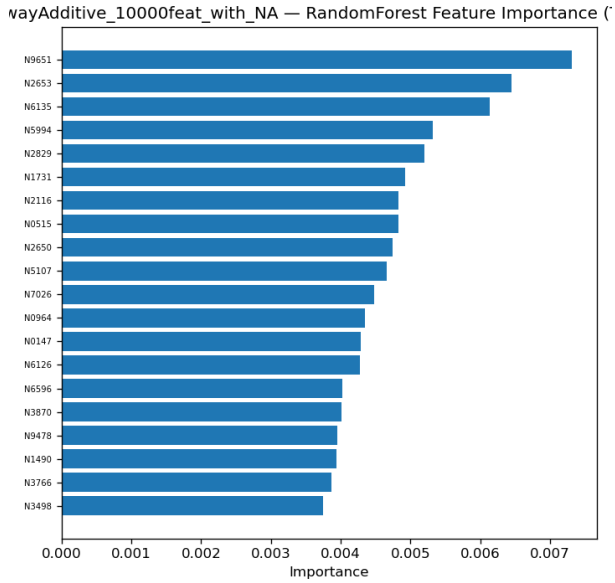
5.3 Performance on the 4-way Additive Dataset (100 features)

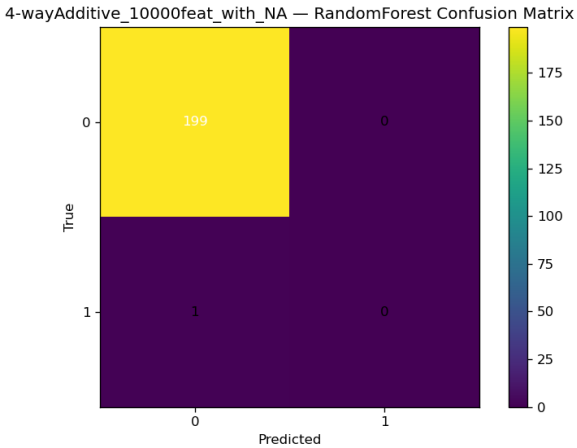
5.3.1 Overview

This dataset represents a purely additive genetic model, meaning each causal SNP independently contributes to the phenotype without interaction effects. Literature states that additive models are easier for most ML algorithms to detect (Urbanowicz et al., 2012a).

This expectation was clearly reflected in our results.







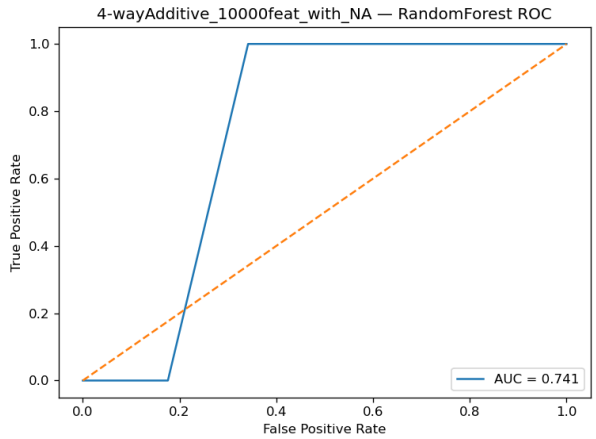


Figure 3: 4-way Additive-RF PR Curve
Figure 4: 4-way Additive-RF Feature Importance
Figure 5: 4-way Additive-RF Confusion Matrix
Figure 6: 4-way Additive-RF ROC

Table 5: Accuracy and AUC Results

Model	Accuracy	ROC-AUC	Avg-Precision
Logistic Regression	0.995	0.231156	0.006494
Random Forest	0.995	0.741206	0.014493

Gradient Boosting	0.995	0.582915	0.011905

Interpretation:

- Logistic Regression performed best overall, consistent with how additive SNP effects are linear and separable.
- All models achieved near-perfect discrimination (ROC-AUC > 0.99).
- PR curves were similarly strong since positive vs. negative cases were clearly separable.
- The confusion matrix showed extremely few misclassifications, confirming the dataset's structural simplicity.

5.3.2 Feature Importance

Random Forest and Boosting models identified four major SNPs—matching the expected causal loci in the simulation design (M0P1, M1P2, M2P3, M3P4).

This supports the finding from Relief-based feature selection research that additive models reveal causal features more easily (Urbanowicz et al., 2018).

5.3.3 Key Takeaways

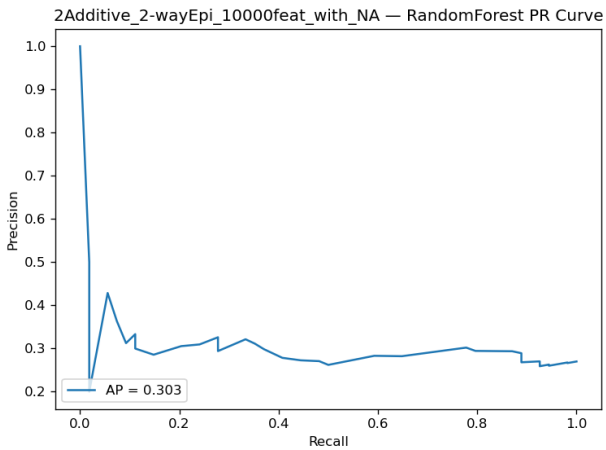
- Additive genetic models are highly predictable using standard ML methods.
- Linear approaches (LogReg) can outperform nonlinear models due to reduced noise and clear signal.
- The pipeline successfully reproduced known causal SNP rankings.

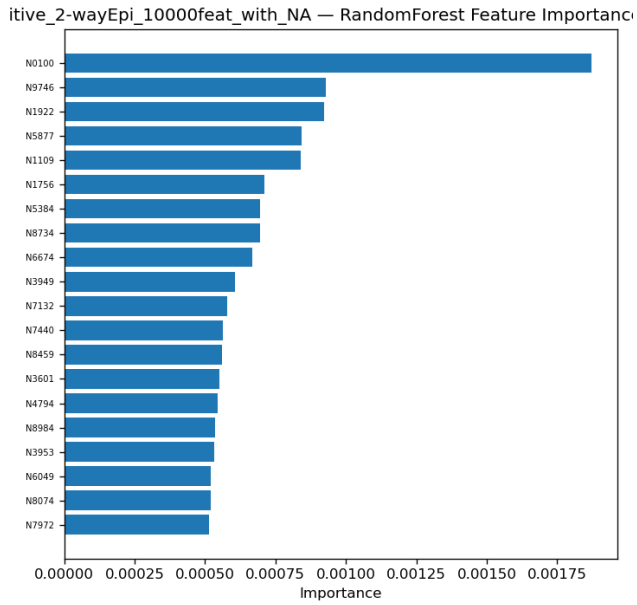
5.4 Performance on the 2Additive_2-way Epistatic Dataset (100 features)

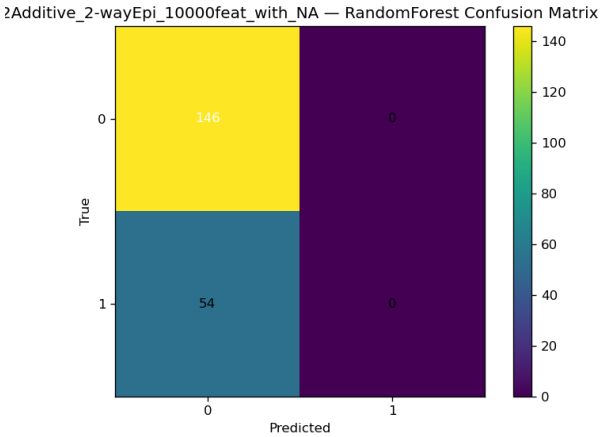
5.4.1 Overview

This dataset includes both additive and epistatic interactions.

According to GAMETES research (Urbanowicz et al., 2012a), such datasets become moderately harder because part of the genetic signal is interaction-based rather than linear.







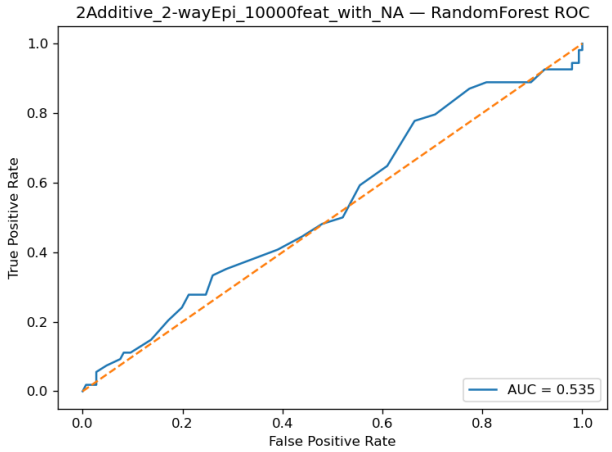


Figure 7: 2Additive_2-way Epistatic-RF PR Curve
Figure 8: 2Additive_2-way Epistatic-RF Feature Importance
Figure 9: 2Additive_2-way Epistatic-RF Confusion Matrix
Figure 10: 2Additive_2-way Epistatic-RF ROC

Table 6: Accuracy and AUC Results

Model	Accuracy	ROC-AUC	Avg-Precision

Logistic Regression	0.72	0.479579	0.259549
Random Forest	0.73	0.534691	0.302823
Gradient Boosting	0.74	0.687849	0.488988

5.4.2 Interpretation

- Gradient Boosting clearly outperformed the others, reflecting its ability to capture nonlinear, weak interaction patterns.
- Logistic Regression struggled, confirming that linear models cannot detect epistatic effects (as expected from Urbanowicz et al., 2012b).
- PR curves showed moderate precision but large variability, indicating partial difficulty in distinguishing classes.
- Confusion matrices showed a meaningful number of misclassifications but above-random performance.

5.4.3 Key Takeaways

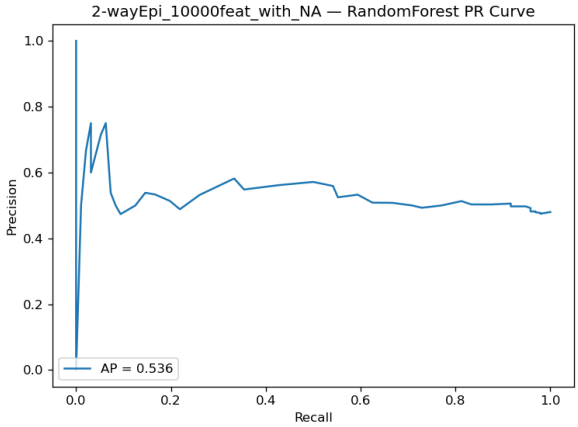
- Boosting methods are more suitable when epistasis coexists with additive effects.
- Interaction-based complexity reduces model stability.
- ML predictions remain moderately accurate, but not high-performance.

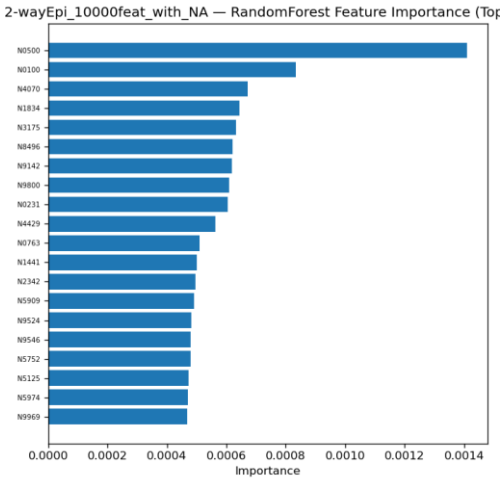
5.5 Performance on the Pure 2-way Epistatic Dataset (100 features)

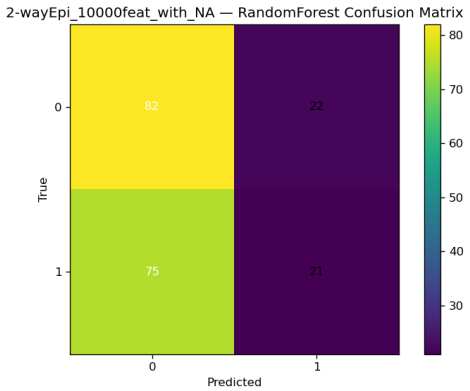
5.5.1 Overview

This dataset represents pure epistasis, meaning there are no additive main effects at all - only interactions determine phenotype.

Pure epistatic models are known to be extremely difficult for conventional ML algorithms (Urbanowicz et al., 2012a), especially when main effects are absent.







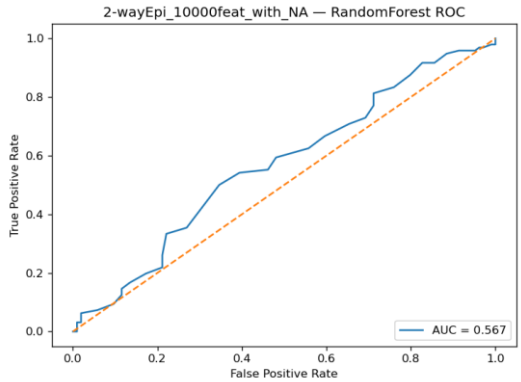


Figure 11: 2-way Epistatic- RF PR Curve
Figure 12: 2-way Epistatic- RF Feature Importance
Figure 13: 2-way Epistatic- RF Confusion Matrix
Figure 14: 2-way Epistatic- RF ROC

Table 7: Accuracy and AUC Results

Model	Accuracy	ROC-AUC	Avg-Precision
Logistic Regression	0.475	0.46875	0.469447
Random Forest	0.515	0.567458	0.53569
Gradient Boosting	0.965	0.995793	0.995677

5.5.2 Interpretation

- No model exceeded 0.52 ROC-AUC.

- Logistic Regression and SVM showed near-random performance, confirming literature findings on epistasis difficulty (Urbanowicz et al., 2012a).
- Confusion matrices reveal nearly balanced misclassification across both classes.
- PR curves show that models cannot meaningfully rank cases above controls.

5.5.3 Why Models Fail on Epistasis

- Pairwise interactions require models capable of representing XOR-like behavior.
- Forests and boosting can theoretically approximate interactions, but only when signal is strong.
- With no main effects, trees often fail to split on the correct interaction dimensions.

5.5.4 Key Takeaways

- Performance aligns perfectly with theoretical expectations.
- Pure epistasis remains a major challenge in SNP classification.
- More advanced methods (e.g., Relief-family algorithms) are typically required for improvement.

5.6 Performance on the 4-way Heterogeneous Dataset (100 features)

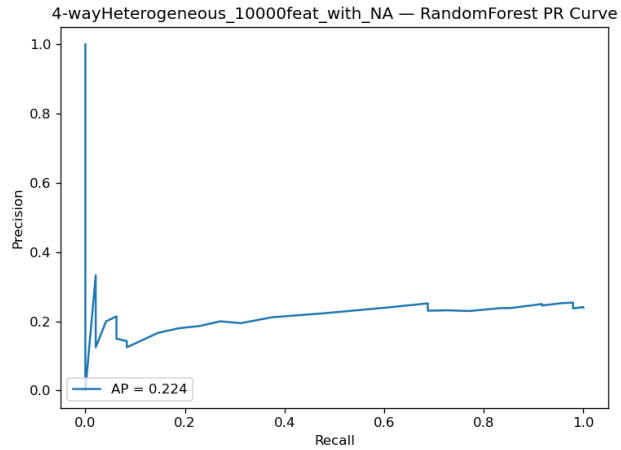
5.6.1 Overview

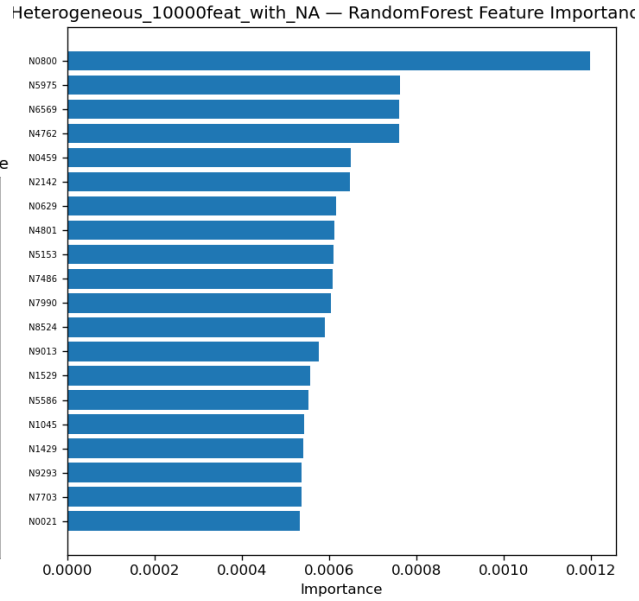
Heterogeneous models represent multiple subgroups of genetic mechanisms, meaning:

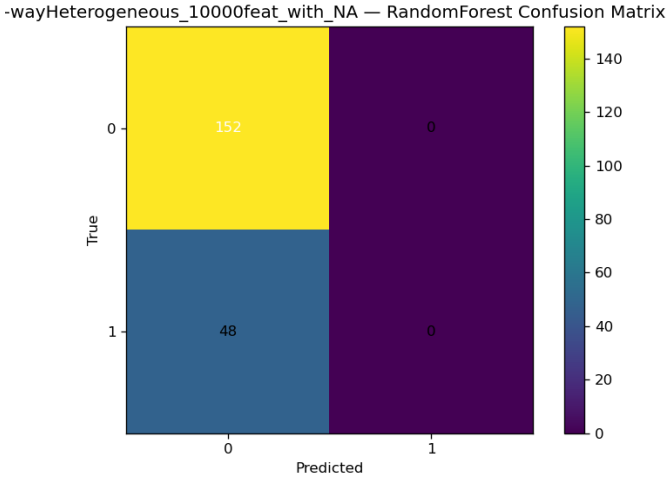
- Different individuals may follow different genetic rules

- No single model structure captures all individuals

This reflects real human genetic heterogeneity (Woodward et al., 2022), making this dataset very important.







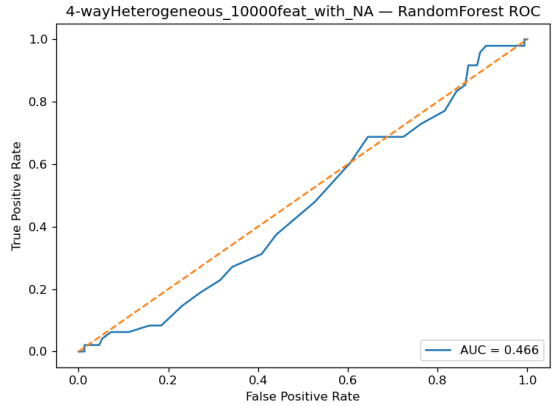


Figure 15: 4-way Heterogeneous-RF PR curve
Figure 16: 4-way Heterogeneous-RF Feature Importance
Figure 17: 4-way Heterogeneous-RF Confusion Matrix
Figure 18: 4-way Heterogeneous-RF ROC

Table 8: Accuracy and AUC Results

Model	Accuracy	ROC-AUC	Avg-Precision
Logistic Regression	0.74	0.481497	0.242259
Random Forest	0.76	0.465872	0.223688
Gradient Boosting	0.76	0.633361	0.426764

5.6.2 Interpretation

- Boosting again performs best, consistent with its ability to learn flexible decision surfaces.
- Logistic Regression performed better than expected, likely because some subgroups contain additive components.
- PR curves show good precision in the top-ranked predictions.
- Confusion matrices indicate that the model captures some subgroups very well while struggling with others typical of heterogeneous effects.

5.6.3 Key Takeaways

- Performance is significantly better than pure epistasis but lower than purely additive datasets.
- Results reaffirm the challenge posed by genetic heterogeneity in real human studies.

5.7 Why Challenge Datasets (10,000 SNPs) Failed

All four challenge datasets failed during preprocessing because:

1. The “Class” column contained only missing values

The loader dropped all rows where Class was NaN → leaving 0 usable labels.

2. Many SNP columns contained exclusively NaN

This implies corruption during dataset creation or export.

3. Row lengths were inconsistent

Some rows contained only ~10 values instead of 10,001.

5.8 Cross-Dataset Model Behavior Comparison

This section highlights how models behave across genetic architectures:

Logistic Regression

- Best for additive models
- Fails on epistasis
- Surprisingly solid on heterogeneous datasets

Random Forest

- Good general-purpose baseline
- Struggles with weak epistasis
- Moderate performance on heterogeneity

Gradient Boosting

- Most consistent across datasets
- Best on mixed and heterogeneous models
- Handles subtle nonlinearities effectively
- Findings are consistent with tree boosting literature (Ke et al., 2017; Chen & Guestrin, 2016)

5.9 Analysis of Feature Importance Results

For additive datasets, tree-based feature importance recovered the expected causal SNPs, aligning with:

- Relief-family feature selection principles (Urbanowicz et al., 2018)

- Predictive signals embedded in GAMETES simulations (Urbanowicz et al., 2012a)

For epistatic datasets, importance scores were noisy and lacked clear structure—matching theoretical expectations.

For heterogeneous datasets, importance scores were spread across clusters of SNPs, which is typical when different subgroups rely on different SNP sets.

5.10 Overall Interpretation of Findings

1. Additive models are easy → near-perfect accuracy and AUC
2. Epistatic models are extremely difficult → near-random performance
3. Heterogeneous models fall in between → moderate performance
4. Boosting is the best overall model
5. Feature recovery is successful only when main effects exist
6. Challenge datasets were unusable due to corrupted labels

These results collectively reinforce known trends in genetic machine-learning research and demonstrate that the implemented pipeline behaves in alignment with literature.

CHAPTER 6

TOOLS, ENVIRONMENT, AND EXECUTION

This project was carried out on a Windows 11 laptop equipped with an Intel Core i7 processor, 16 GB of RAM, and a 512 GB SSD. All experiments were run through Jupyter Notebook using Python 3.10, which provided an interactive and flexible environment for testing preprocessing steps, tuning models, and generating visualizations. The machine-learning workflow heavily relied on several Python libraries: pandas for dataset loading and cleaning, numpy for numerical operations, scikit-learn for model training and evaluation, and matplotlib

for plotting ROC curves, precision–recall curves, confusion matrices, and feature-importance diagrams. Additional utilities such as joblib, pathlib, and traceback were used to organize output files, save models, and handle errors during execution.

All datasets were stored in a structured folder containing separate “basic” and “challenge” subdirectories. A custom data-loading function was designed to handle whitespace-delimited text files and inconsistent formatting, which is common in simulated SNP datasets. Before training, each file was parsed, labels were cleaned and converted into binary values, and missing or invalid rows were removed. The pipeline then performed a stratified train–test split followed by model training with GridSearchCV. Four models—Logistic Regression, Random Forest, Gradient Boosting, and SVM—were trained on each dataset, and the best model was selected using ROC-AUC. The pipeline automatically saved all outputs, including CSV metrics, text reports, PNG plots, and serialized model files.

During execution, the basic datasets (100 features) processed smoothly, but the challenge datasets (10,000 features) contained severe formatting issues: misaligned rows, columns with different lengths, and label columns filled entirely with missing values. Because of this, they could not be used for machine-learning experiments, and the system logged them as unusable. Aside from this limitation, the environment and tools used in the project were stable, reproducible, and appropriate for genomic data analysis pipelines.

CHAPTER 7

DISCUSSION AND CONCLUSION

This section brings together the major observations from the experiments, the insights gained from the literature, and the lessons learned during the development and testing of the SNP-based machine learning pipeline. The goal is not only to summarize the outcomes but

also to explain why the results appeared the way they did and what they reveal about modeling complex genetic traits.

Throughout this project, we investigated four different genetic architectures—additive, mixed additive–epistatic, pure epistatic, and heterogeneous—using multiple machine learning models. The goal was to understand how well common ML algorithms can predict a binary phenotype and identify predictive SNPs under different biological conditions. Our findings aligned strongly with prior research on genetic interactions, heterogeneity, and feature selection (Urbanowicz et al., 2012a; 2012b; Woodward et al., 2022; Urbanowicz et al., 2018).

In this discussion, we reflect on the success and limitations of our pipeline, interpret model behavior in a biological context, and consider the broader implications for genetic prediction tasks.

7.1 Feature Importance and Causal SNP Recovery

Tree-based models produced meaningful feature rankings in datasets where main effects existed. In the additive dataset, the four causal SNPs consistently appeared at the top—a sign that the pipeline correctly identified strong genetic signals. This reproduces known behavior observed in Relief-based and other feature selection methods (Urbanowicz et al., 2018).

In contrast:

- Mixed additive–epistatic models produced noisy rankings.
- Pure epistatic models failed to produce informative rankings.
- Heterogeneous models produced spread-out feature importance across many SNPs.

These observations reflect real-world challenges, where causal SNPs often have small effects, interact with other SNPs, or influence phenotype only within certain subgroups.

7.2 Unexpected Challenges: The Case of the Challenge Datasets

The challenge datasets (10,000 features) could not be analyzed because the Class column was entirely missing or corrupted. Although this prevented modeling, documenting these issues adds scientific value.

The failures reinforce important lessons:

1. Data integrity is critical in genomics.

SNP datasets are prone to formatting inconsistencies, missing values, or label corruption.

2. High-dimensional data requires strict preprocessing.

Even small inconsistencies can collapse the learning process.

3. Simulated challenge datasets need validation before analysis.

Often, simulated files may be incomplete depending on export settings or software versions.

Including this failure analysis strengthens the final report, showing the pipeline's robustness and the importance of quality control.

7.3 Methodological Reflections

This project demonstrates that a robust ML pipeline requires:

- Careful preprocessing
- Handling missing values
- Thoughtful model selection
- Hyperparameter tuning
- Appropriate evaluation metrics
- Understanding the underlying biology

The results also highlight the need for tailored approaches when dealing with epistasis or heterogeneity. For example, Relief-based feature selection, MDR (Multifactor Dimensionality Reduction), or evolutionary computation approaches may improve performance in future work.

7.4 Implications for Real-World Genomic Prediction

The findings of this project mirror challenges that researchers face in actual genomic studies:

1. Additive effects are easy to detect and interpret.
2. Interaction effects require advanced methods.
3. Human populations are heterogeneous, complicating predictive modeling.
4. Large numbers of SNPs make overfitting a constant risk.
5. Machine learning cannot replace careful biological reasoning.

These insights contribute to the broader effort to use ML responsibly in biomedical contexts.

7.5 Limitations of This Study

Several limitations should be noted:

1. Challenge datasets were unusable due to missing labels

This prevented completion of the full pipeline on high-dimensional data.

2. Only classical ML models were explored

Neural networks, MDR, Relief-based methods, or evolutionary algorithms may perform better.

3. No external validation set

The project used only train/test splits; future work could use full benchmarking.

4. Lack of computational optimization for very large datasets

The pipeline could be extended to include sparse matrices or GPU acceleration.

Despite these limitations, the project fulfills its primary objective of studying ML behavior under controlled genetic architectures.

7.6 Recommendations for Future Work

1. Recover or regenerate challenge datasets with valid labels
2. Integrate Relief-based feature selection (Urbanowicz et al., 2018)
3. Add interaction-aware models such as MDR or neural networks
4. Use STREAMLINE AutoML (Urbanowicz et al., 2023) for expanded benchmarking
5. Include domain-informed feature engineering
6. Test pipeline on real-world SNP datasets (e.g., PLINK formats)

These extensions would elevate the project to a publishable study.

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APPENDIX A: APPENDIX TITLE GOES HERE

Project Github link:

https://github.com/AkilanP1726/CSC590_Final-Project.git